David G Courtney BSc PhD

Senior Lecturer in Molecular Virology, Queen's University Belfast
UK Ambassador for the European Research Council

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Molecular virologist and RNA biologist. David leads The Courtney Group at Queen's University Belfast, which studies how influenza A virus RNA recruits host machinery to be modified, exported, translated and trafficked, and how those dependencies can be turned into RNA therapeutics and antivirals. His work on mRNA modifications such as m6A, m5C and pseudouridine, which control how well an mRNA is translated, bears directly on mRNA vaccines and medicines. He has won over €1.8M in competitive funding as sole PI, sits on BBSRC and European funding panels, and represents the United Kingdom as an Ambassador for the European Research Council.

1,447 citations · h-index 19 · 42 indexed works (OpenAlex, 2026-11-09)

APPOINTMENTS

2025 – **Senior Lecturer in Molecular Virology**, Queen's University Belfast, UK

2021 – 2025 **Lecturer and ERC Research Fellow**, Queen's University Belfast, UK
Founded The Courtney Group with an ERC Starting Grant.

2020 – 2021 **Postdoctoral Research Fellow in Molecular Virology**, Queen's University Belfast, UK
High-throughput drug repurposing screens against SARS-CoV-2 in airway epithelial cells.

2019 – 2020 **Marie Skłodowska-Curie Postdoctoral Fellow**, Institut Pasteur, Paris, France
Developed methods to identify the host RNA-binding proteins of influenza A virus RNAs.

2016 – 2019 **Marie Skłodowska-Curie Postdoctoral Fellow**, Duke University, North Carolina, USA
Showed how m6A and m5C RNA modifications regulate influenza A virus, HIV-1 and other viruses.

2010 – 2011 **Industrial placement, BioPharm R&D**, GlaxoSmithKline, UK
Twelve months in molecular expression and cell line development.

EDUCATION

2012 – 2015 **PhD, Molecular Biology**, Ulster University, UK
The development of personalised medicines for corneal dystrophies (siRNA and CRISPR therapeutics).

2008 – 2012 **BSc (Hons) Biomedical Sciences with industrial studies, First Class**, Ulster University, UK

LEADERSHIP AND SERVICE

2026 – **Entrepreneurial Lead, Innovate UK ICURe Discover programme**, Innovate UK, UK Research and Innovation
Competitively selected for the UKRI-funded programme that helps researchers test the commercial potential of their research through market discovery.

2025 – **UK Ambassador for the European Research Council**, Ambassadors for the ERC network
One of 32 ERC grantees chosen to make the case for frontier research to policymakers, the media and the research community.

Current **Vice-Chair, Marie Skłodowska-Curie Actions Postdoctoral Fellowships panel**, European Commission

2024 – **Member, BBSRC Panel of Experts**, UK Research and Innovation

Current **Elected member, Virology Division Committee**, Microbiology Society

2024 **Chair and organiser, UK ERC Showcase**, Queen's University Belfast
Flagship event with panels featuring Nobel Laureate Prof Emmanuelle Charpentier and Prof Teresa Lambe OBE.

2023 – 2024 **Contributor, US National Academies report "Toward Sequencing and Mapping of RNA Modifications"**, US National Academy of Sciences
Invited ideation panel delegate (2023) and co-author of a commissioned paper (2024).

2021 – 2024 **Grant reviewer**, MSCA (reviewer and rapporteur), US National Science Foundation panel, Swiss National Science Foundation, Polish National Science Centre

Current **Module Coordinator, MSc Research Project**, Wellcome-Wolfson Institute for Experimental Medicine, Queen's

Current **STEM NI Ambassador**, Northern Ireland

2014 – **Fellow of the Winston Churchill Memorial Trust**, Churchill Fellowship, UK

GRANTS AND AWARDS

2021 – 2026 **ERC Starting Grant: PTFLU, post-transcriptional regulation of influenza A virus RNA**, European Research Council · sole PI · grant 949506
€1,568,010 to establish The Courtney Group.

2023 – 2026 **MRC research grant: host ATPases in rhinovirus replication**, UKRI Medical Research Council · co-investigator · MR/X020371/1
£458,574.

2024 – 2025 **Royal Society Partnership Grant**, The Royal Society · with Regent House School
A research project run with secondary school pupils.

2017 – 2020 **Marie Skłodowska-Curie Global Fellowship: IAV-m6A**, European Commission · sole PI · Duke University and Institut Pasteur
€260,000 to study m6A RNA methylation in influenza A virus replication and pathogenesis.

2015 **Postgraduate Student of the Year (John RE Scott Convocation award)**, Ulster University

2014 – 2015 **Churchill Fellowship**, Winston Churchill Memorial Trust
€7,600 for research stays in Italy and Denmark.

PATENT APPLICATIONS

2024 **Novel compounds for the treatment of viral infections**, US patent application 18/550,005

2019 **Compositions and methods for enhanced gene expression and viral replication**, US patent application 16/091,822
Using m6A-associated cellular machinery to boost transgene expression or viral replication.

2019 **Single guide RNA/CRISPR/Cas9 systems, and methods of use thereof**, US patent application 16/326,908
Allele-specific CRISPR/Cas9 systems for treating corneal dystrophies.

SELECTED TALKS AND EVENTS

2026 **Invited speaker: Building ERC-competitive research**, University of Split, Croatia

2024 **Event chair: UK ERC Showcase**, Queen's University Belfast

2023 **Delegate: ideation panel on RNA modification mapping**, US National Academy of Sciences

2020 **Invited speaker: ANRS International Symposium on RNA Modifications**, Institut Cochin, Paris

MENTORING AND LEADERSHIP

- Mentored Dr Hannah Turkington to a competitive three-year BBSRC Postdoctoral Fellowship (2024).
- PhD graduates went straight to postdoctoral posts at the University of Cambridge and the University of Copenhagen.
- MSc graduates have moved into government science roles, including Forensic Science NI.
- EMBO Laboratory Leadership course and UKRI Future Leaders Fellows leadership retreat (2024).

PUBLIC ENGAGEMENT AND MEDIA

- Regular expert voice on BBC television and national radio during the COVID-19 pandemic, explaining viral mutation, immune escape and variant surveillance.
- Long-form articles for The Conversation reaching international readers.
- Works with regional health bodies and schools to promote influenza vaccination awareness.

PROFESSIONAL MEMBERSHIPS

Microbiology Society · American Society for Virology · RNA Society

SELECTED PUBLICATIONS

co-senior author · * co-first author. Full list at davidgcourtney.com/publications and on ORCID.

1. Bonazza S, Turkington HL, Sukumar S, Peate E, Coutts HL, Montgomery JJ, Hawthorn C, Getty EMPE, Touzelet O, Barabas J, Power UF, **Courtney DG** (2026). Myoferlin is a component of late-stage vRNP trafficking vesicles for enveloped RNA viruses. *Nature Communications 17*. Cited 3.
2. Bonazza S, **Courtney DG** (2025). Influenza A virus RNA localisation and the interceding trafficking pathways of the host cell. *PLoS Pathogens 21(4): e1013090*. Cited 13.

3. Bonazza S, Coutts HL, Sukumar S, Turkington HL, **Courtney DG** (2024). Identifying cellular RNA-binding proteins during infection uncovers a role for MKRN2 in influenza mRNA trafficking. *PLoS Pathogens 20(5): e1012231*. Cited 11.
4. Dupont M, Krischuns T, Giai-Gianetto Q, Paisant S, Bonazza S, Brault J, Douché T, Perez-Perri JI, Hentze MW, Cusack S, Matondo M, Isel C, **Courtney DG#**, Naffakh N# (2024). The RBPome of influenza A virus mRNA reveals a role for TDP-43 in viral replication. *Nucleic Acids Research*. Cited 17.
5. Coutts LH, **Courtney DG** (2024). Identifying individual pseudouridine (Ψ) sites across transcripts from HIV-1 infected cells. *HIV Protocols. Methods in Molecular Biology, vol 2807*.
6. **Courtney DG**, Tsai K, Bogerd HP, et al. (2019). Epitranscriptomic regulation of HIV-1 gene expression by m5C. *Cell Host & Microbe 26(2): 217–227*. Cited 226.
7. **Courtney DG**, Chalem A, Bogerd HP, et al. (2019). Extensive epitranscriptomic methylation of A and C residues on murine leukemia virus transcripts enhances viral gene expression. *mBio 10(3): e01209-19*. Cited 76.
8. **Courtney DG**, Kennedy EM, Dumm RE, et al. (2017). Epitranscriptomic enhancement of influenza A virus gene expression and replication. *Cell Host & Microbe 22(3): 377–386*. Cited 232.
9. **Courtney DG**, Moore JE, Atkinson SD, et al. (2015). CRISPR/Cas9 DNA cleavage at SNP-derived PAM enables both in vitro and in vivo KRT12 mutation-specific targeting. *Gene Therapy 23(1): 108–112*. Cited 125.
10. **Courtney DG**, Toftgaard Poulsen E, Kennedy S, et al. (2015). Protein composition of TGFBI-R124C- and TGFBI-R555W-associated aggregates suggests multiple mechanisms leading to lattice and granular corneal dystrophy. *Investigative Ophthalmology & Visual Science 56: 4653–4661*. Cited 38.